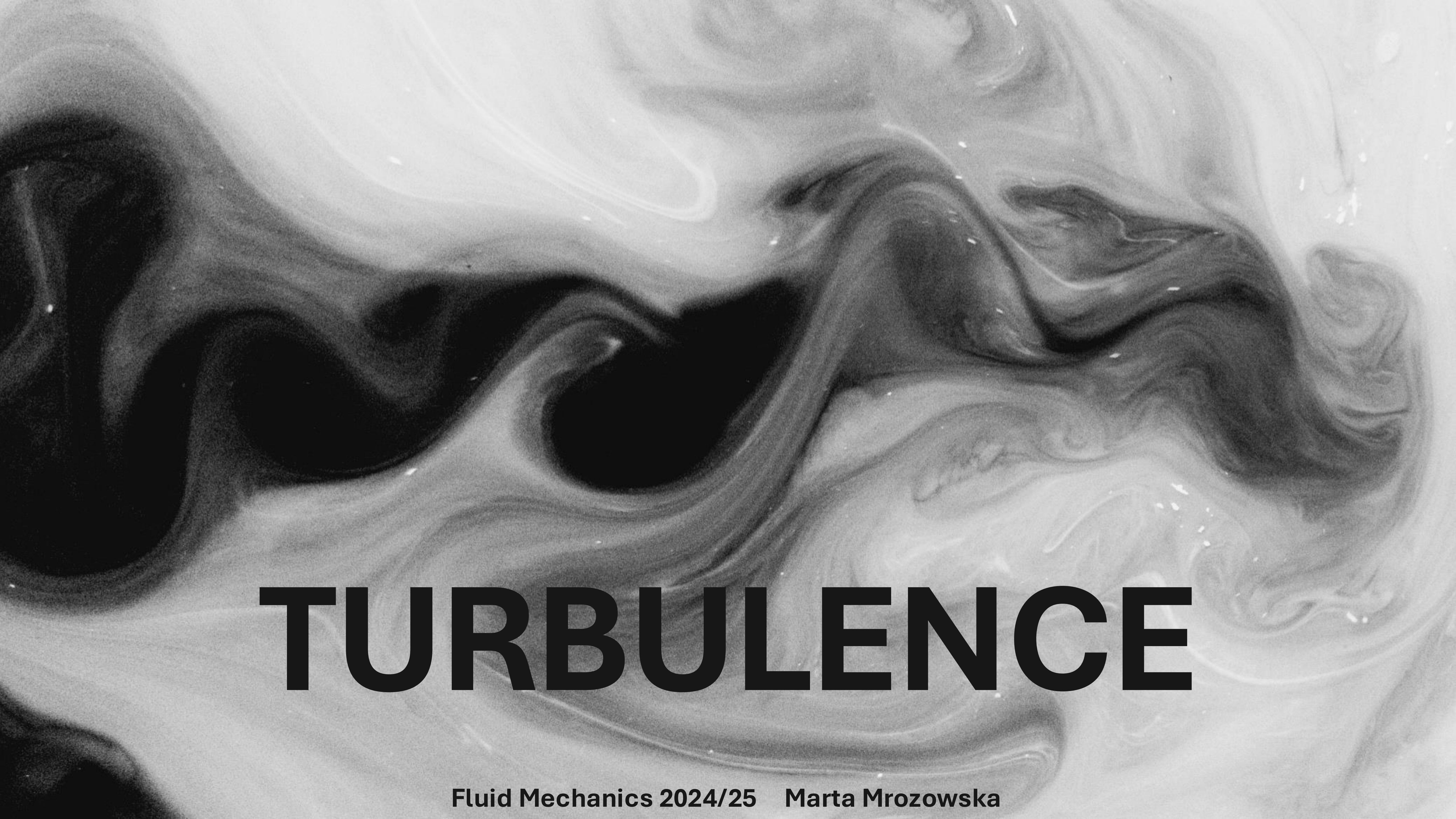
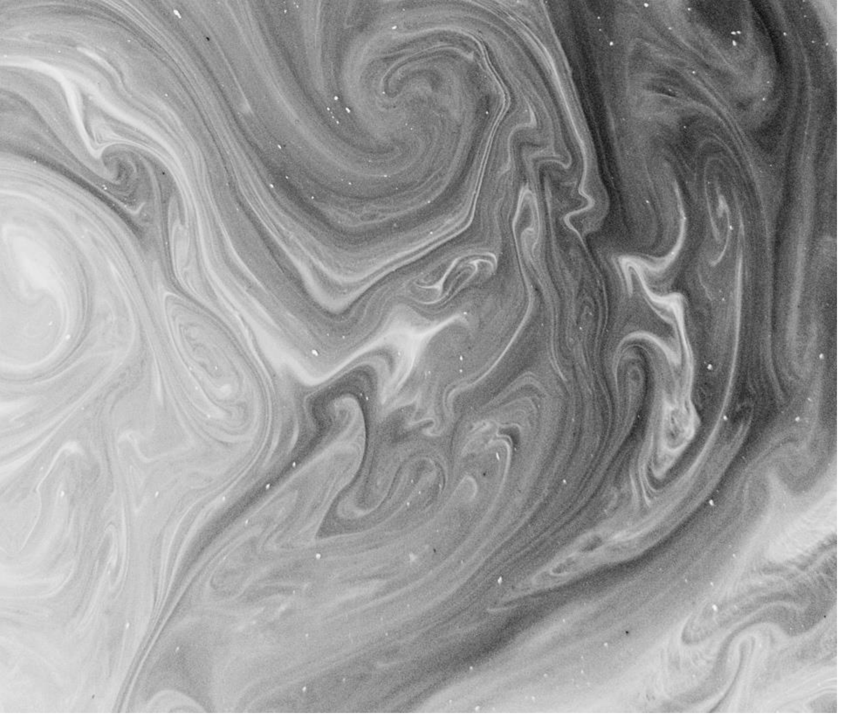
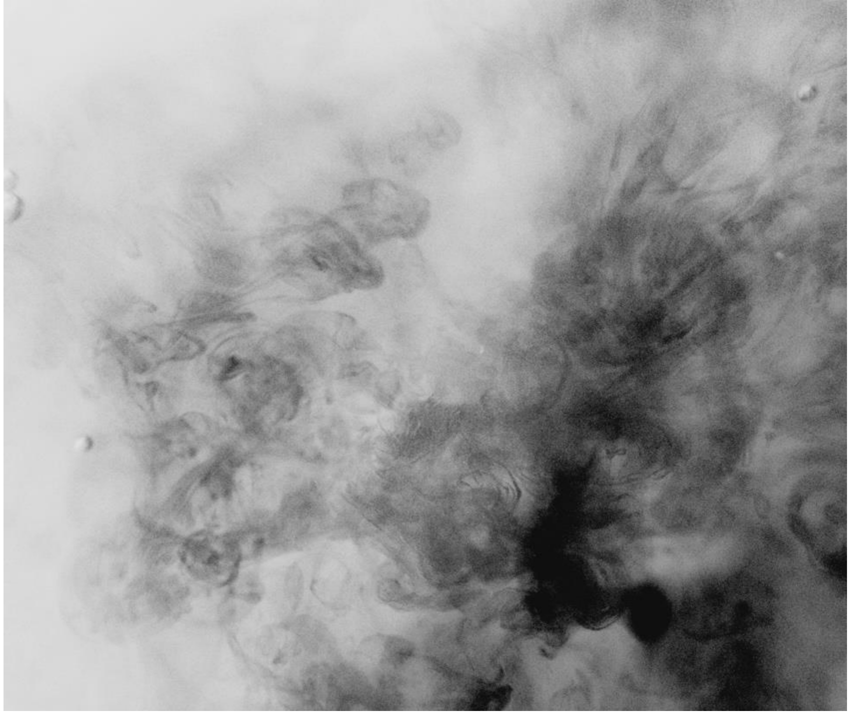
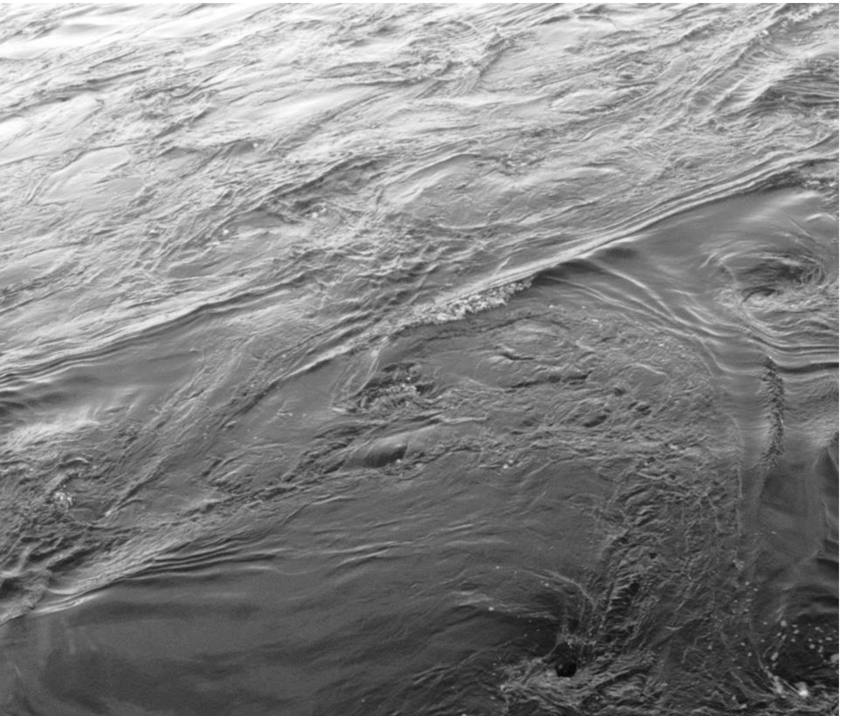
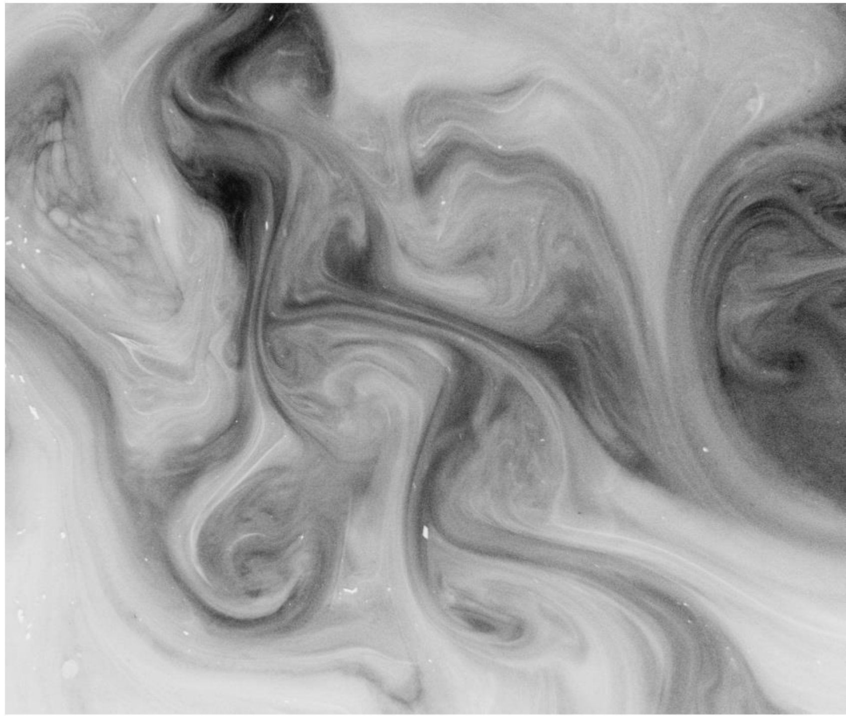


TURBULENCE

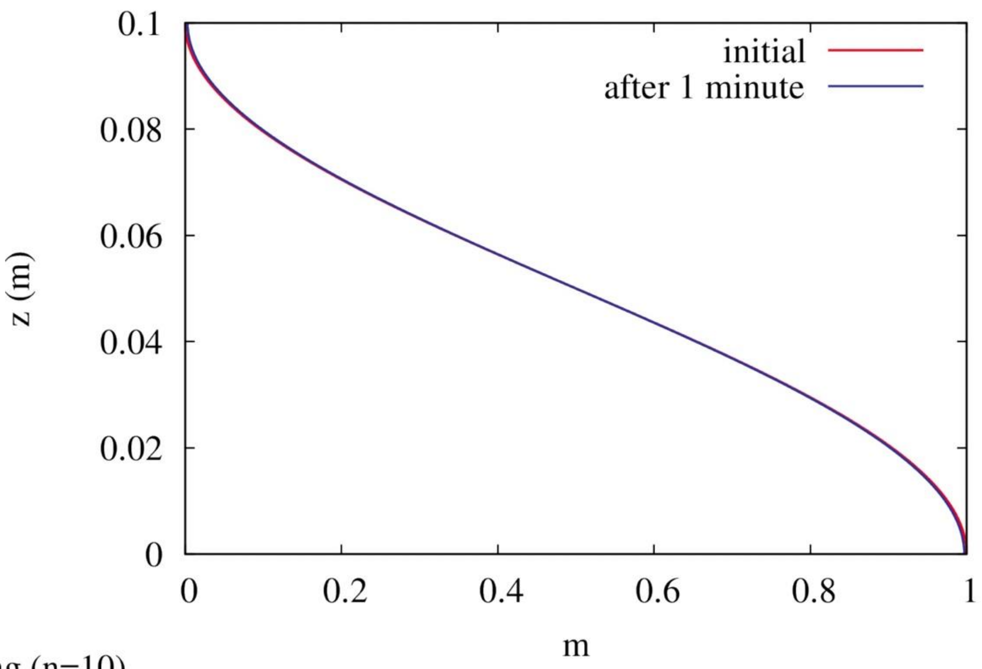
Fluid Mechanics 2024/25 Marta Mrozowska

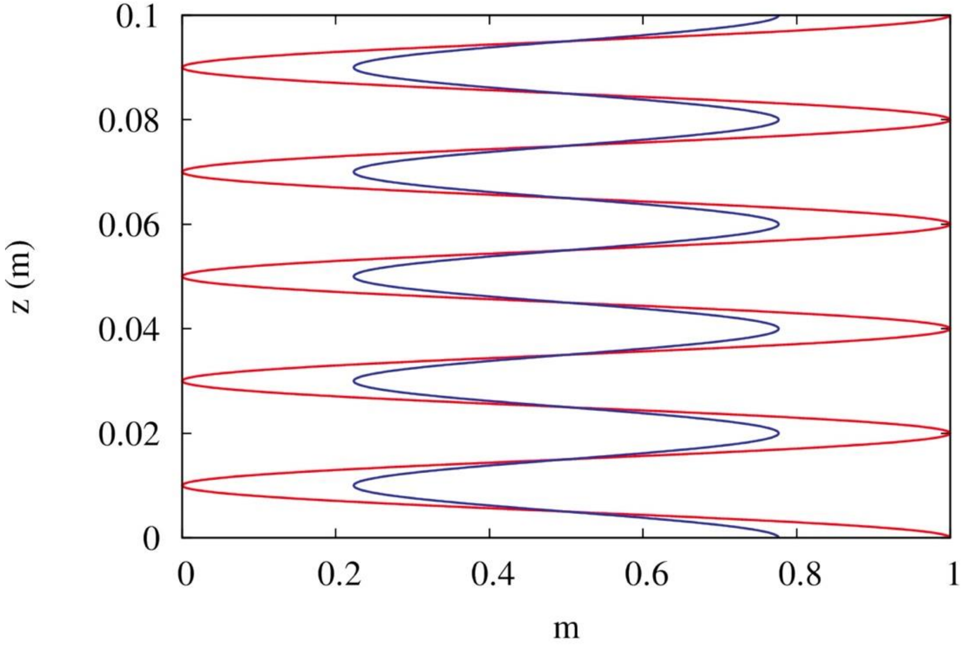




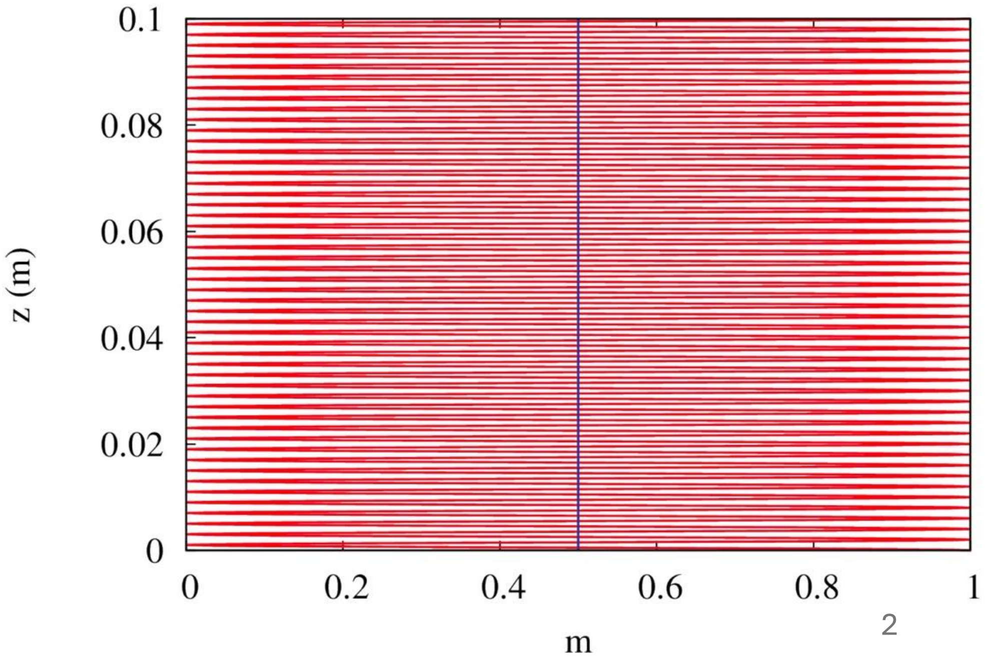
no stirring (n=1)



moderate stirring (n=10)



strong stirring (n=100)



Turbulence…

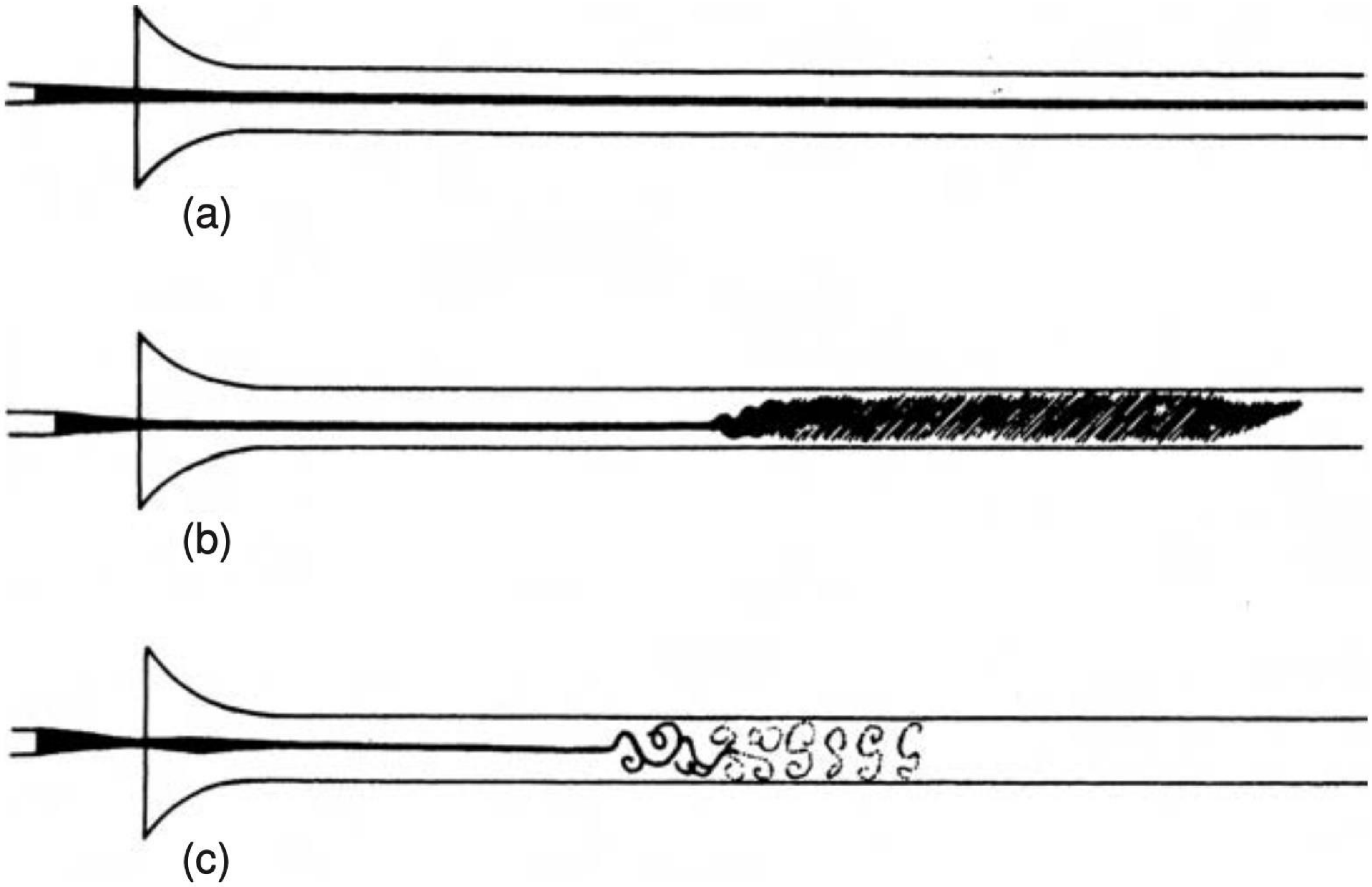
…contains vortex-like structures (eddies).
…is an efficient mixer.
…is chaotic and random.
…dissipates momentum.

LEARNING GOALS

- Get introduced to the concept of **energy cascade**
- Understand **Kolmogorov’s three hypotheses**
- Transition from **physical** to **wavenumber space**
- Derive the **Kolmogorov spectrum**

NOT LEARNING GOALS

- X Understand turbulence
- X Understand the formal mathematical arguments behind Kolmogorov’s three hypotheses
- X Derive the transition between the physical and wavenumber spectrum



Reynolds (1893), Thorpe (2007)

IV. *On the Dynamical Theory of Incompressible Viscous Fluids and the Determination of the Criterion.*

By OSBORNE REYNOLDS, *M.A., LL.D., F.R.S., Professor of Engineering in Owens College, Manchester.*

Received April 25—Read May 24, 1894.

2. In 1883 I succeeded in proving, by means of experiments with colour bands—the results of which were communicated to the Society*—that when water is caused by pressure to flow through a uniform smooth pipe, the motion of the water is *direct*, *i.e.*, parallel to the sides of the pipe, or *sinuous*, *i.e.*, crossing and re-crossing the pipe, according as U_m, the mean velocity of the water, as measured by dividing Q, the discharge, by Δ, the area of the section of the pipe, is below or above a certain value given by

$$K\mu/D\rho,$$

where D is the diameter of the pipe, ρ the density of the water, and K a numerical constant, the value of which according to my experiments and, as I was able to show, to all the experiments by POISEUILLE and DARCY, is for pipes of circular section between

$$1900 \text{ and } 2000,$$

or, in other words, steady direct motion in round tubes is stable or unstable according as

$$\rho\frac{DU_m}{\mu} < 1900 \text{ or } > 2000,$$

the number K being thus a criterion of the possible maintenance of sinuous or eddying motion.

Reynolds (1894)

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1900 and 2000,

or, in other words, steady direct motion in round tubes is stable or unstable according as

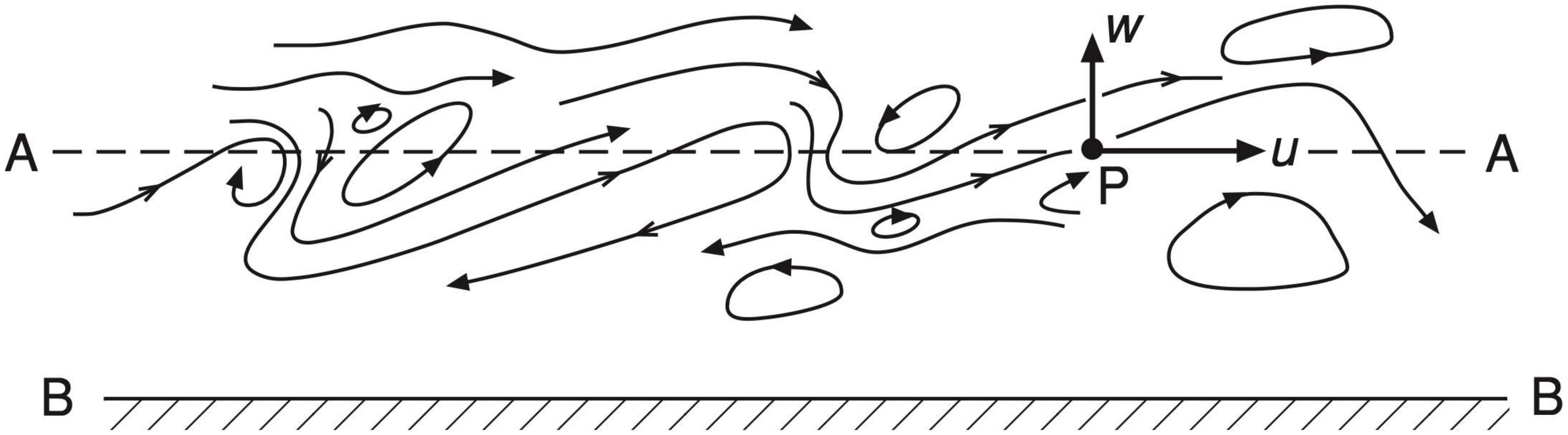
$$\mathrm{Re} = \frac{\mathcal{L}\mathcal{U}}{\nu} \quad \rho\frac{DU_m}{\mu} < 1900 \text{ or } > 2000,$$

the number K being thus a criterion of the possible maintenance of sinuous or eddying motion.

Reynolds (1894)

Reynolds decomposition: $u = \overline{u} + u'$

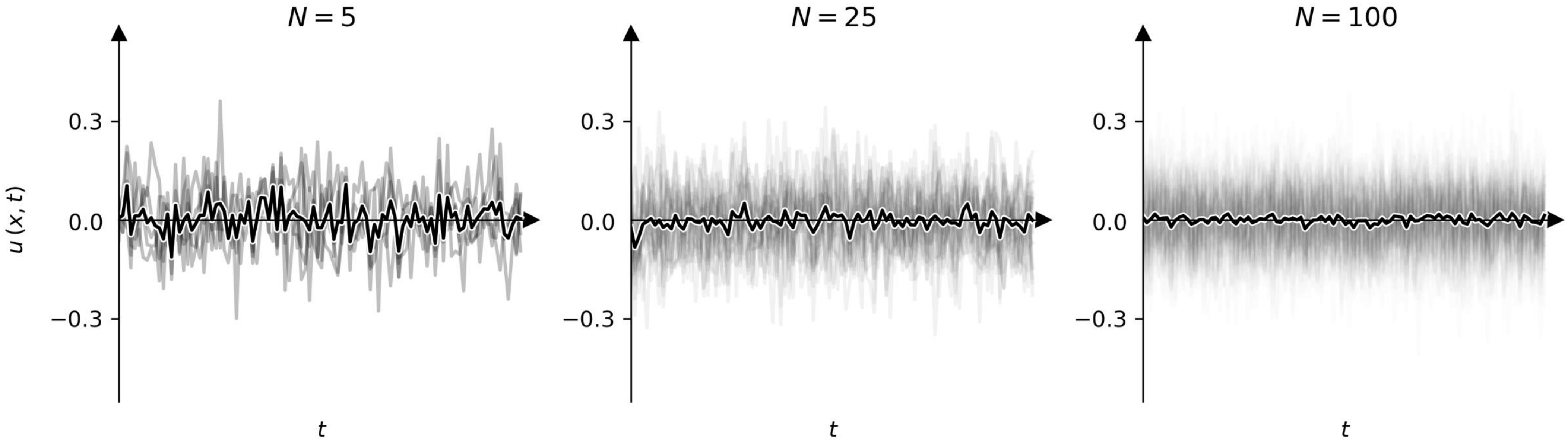
Reynolds stress tensor: $\overline{u'_i u'_j}$



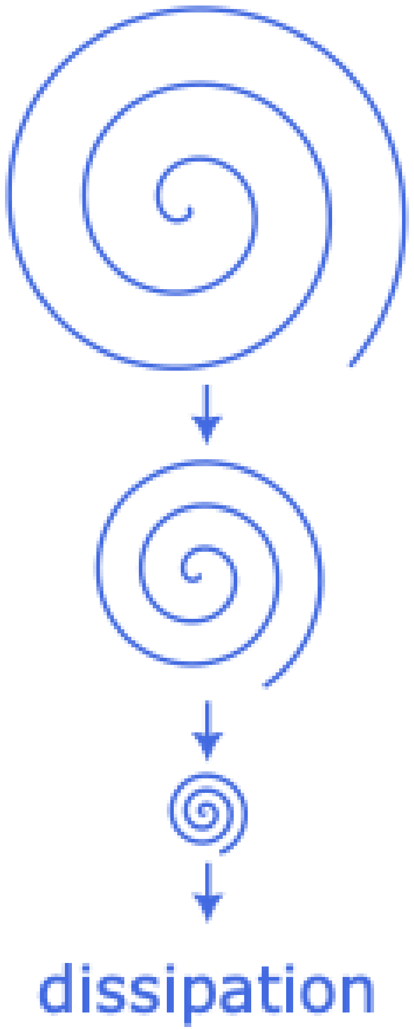
Thorpe (2007)

In turbulent flow, $u(x,t)$ is *random*.

$$\langle u(x,t)\rangle = \lim_{N\to\infty} \overline{u(x,t)} = \lim_{N\to\infty} \frac{1}{N}\sum_{n=1}^{N} u(x,t\!:\!n)$$

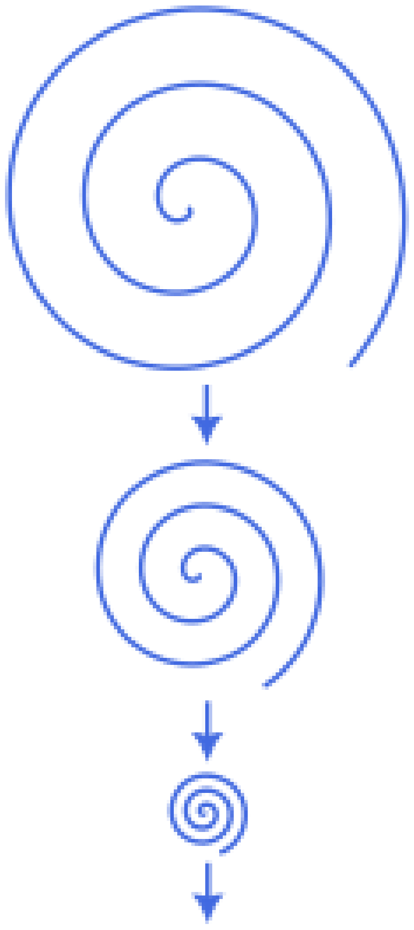


THE ENERGY CASCADE



Big whorls have little whorls,
Which feed on their velocity;
And little whorls have lesser whorls,
And so on to viscosity
(in the molecular sense).

Richardson (1922), Pope (2000)



Kinetic energy is **dissipated** at scales much smaller than the scale at which it **enters** the system.

It is transferred between scales via **inviscid** processes.

Dissipation scales: l_k, t_k, v_k

Production scales: l_0, t_0, v_0

Assumption:

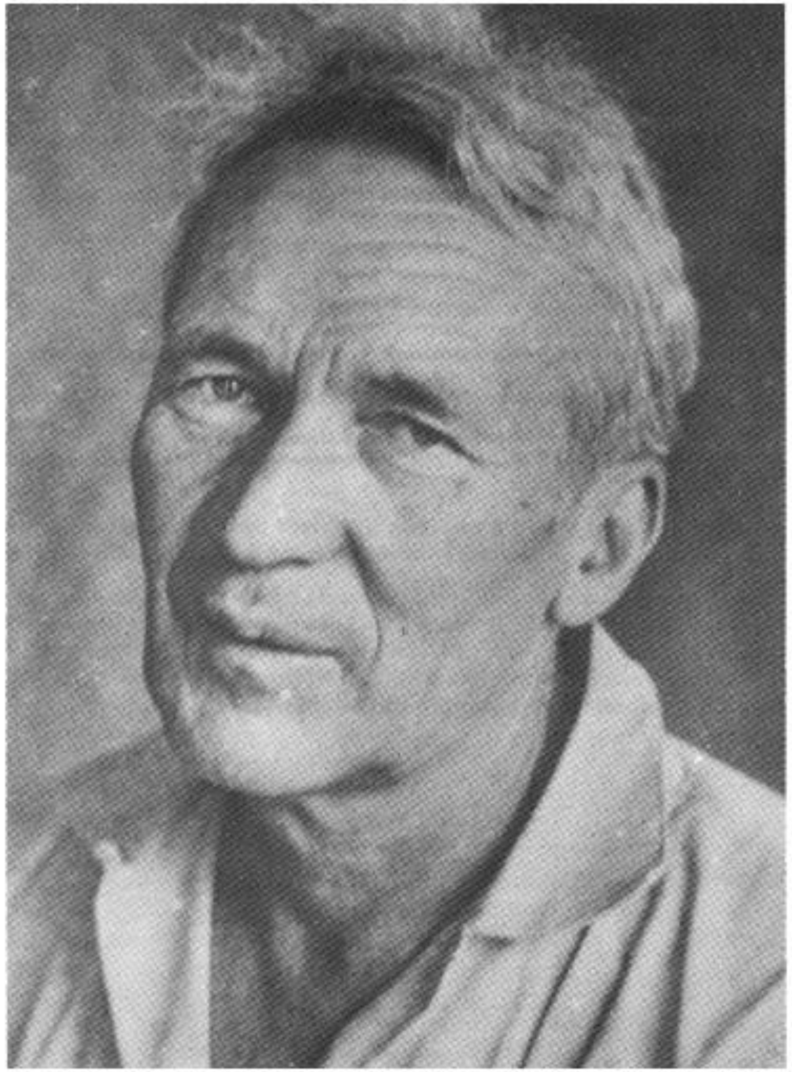
$\frac{l_k}{l_0} \ll 1$ $\frac{t_k}{t_0} \ll 1$ $\frac{v_k}{v_0} \ll 1$

Summary I:

The ”end points” of the energy cascade are *observable* in the sense that turbulent kinetic energy production and dissipation rates can be measured.

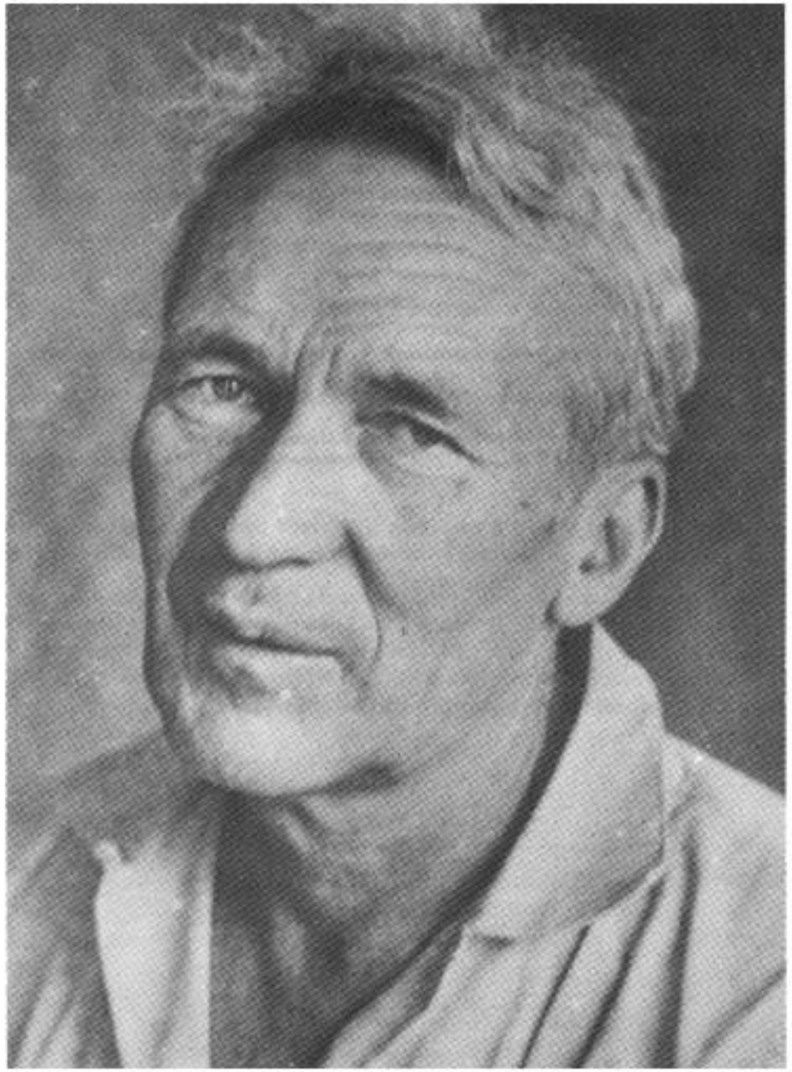
The energy transfer from large to small scales by inviscid processes is a *hypothesis*.

KOLMOGOROV'S HYPOTHESES

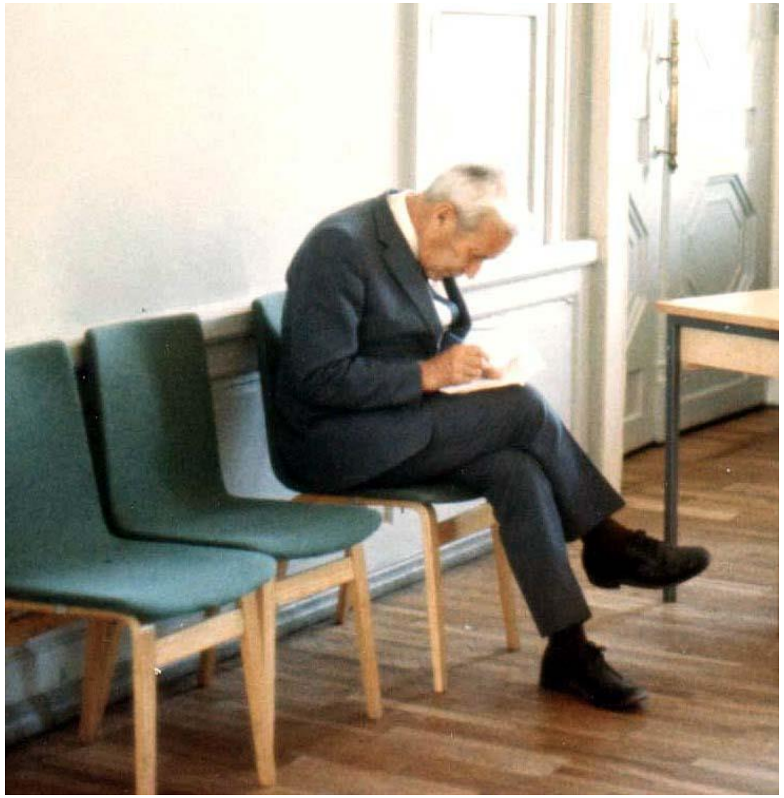


Andrei Nikolaevich Kolmogorov, 25 April 1903–20 October 1987

https://doi.org/10.1017/S0021900200041115 Published online by Cambridge University Press



Andrei Nikolaevich Kolmogorov, 25 April 1903–20 October 1987



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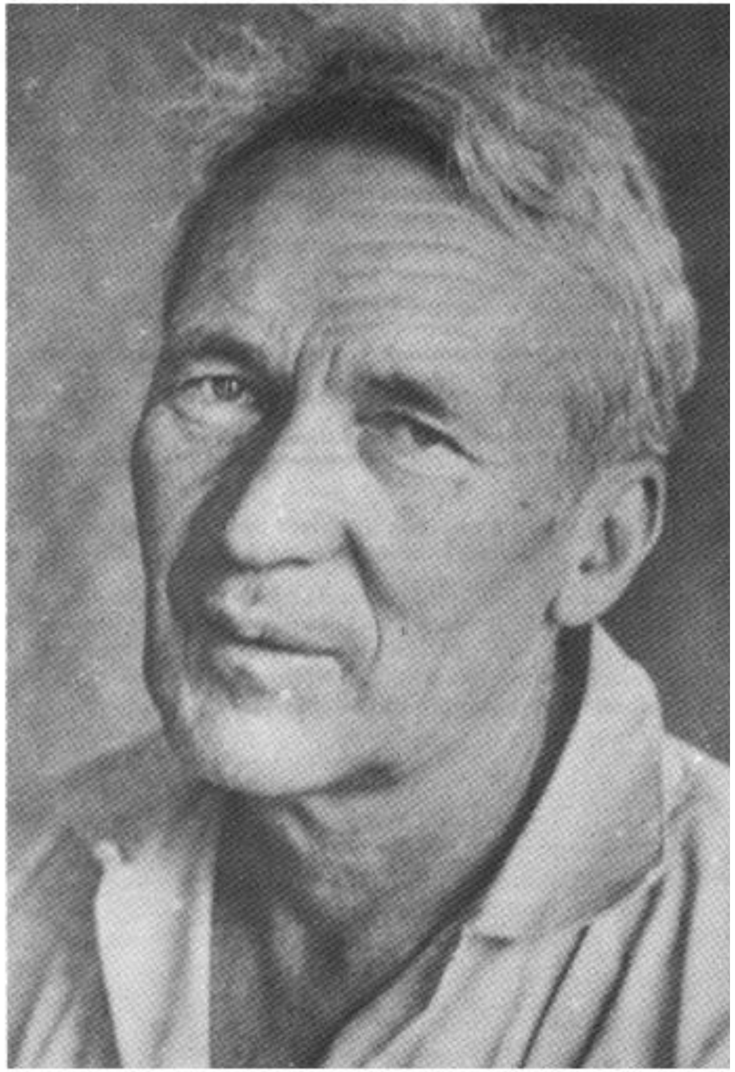

Andrei Nikolaevich Kolmogorov, 25 April 1903–20 October 1987

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- Kolmogorov microscales
- Kolmogorov's normability criterion
- Fréchet–Kolmogorov theorem
- Kolmogorov space
- Kolmogorov complexity
- Kolmogorov–Smirnov test
- Wiener filter (also known as Wiener–Kolmogorov filtering theory)
- Wiener–Kolmogorov prediction
- Kolmogorov automorphism
- Kolmogorov's characterization of reversible diffusions
- Borel–Kolmogorov paradox
- Chapman–Kolmogorov equation
- Hahn–Kolmogorov theorem
- Johnson–Mehl–Avrami–Kolmogorov equation
- Kolmogorov–Sinai entropy
- Astronomical seeing described by Kolmogorov's turbulence law
- Kolmogorov structure function
- Kolmogorov–Uspenskii machine model
- Kolmogorov's zero–one law
- Kolmogorov–Zurbenko filter
- Kolmogorov's two-series theorem
- Rao–Blackwell–Kolmogorov theorem
- Khinchin–Kolmogorov theorem
- Kolmogorov population model
- Kolmogorov's Strong Law of Large Numbers

- Fisher–Kolmogorov equation
- Johnson–Mehl–Avrami–Kolmogorov equation
- Kolmogorov axioms
- Kolmogorov equations (also known as the Fokker-
- Kolmogorov dimension (upper box dimension)
- Kolmogorov–Arnold theorem
- Kolmogorov–Arnold–Moser theorem
- Kolmogorov continuity theorem
- Kolmogorov's criterion
- Kolmogorov extension theorem
- Kolmogorov's three-series theorem
- Convergence of Fourier series
- Gnedenko-Kolmogorov central limit theorem
- Quasi-arithmetic mean (it is also called Kolmogoro
- Kolmogorov homology
- Kolmogorov's inequality
- Landau–Kolmogorov inequality
- Kolmogorov integral
- Brouwer–Heyting–Kolmogorov interpretation

$$\frac{\partial \bar{e}}{\partial t} + \overline{u_j}\frac{\partial \bar{e}}{\partial x_j} = \frac{\partial}{\partial x_j}\left(-\frac{1}{2}\overline{u'_i u'_i u'_j} + 2\nu\overline{u'_i s'_{ij}} - \frac{\overline{p' u'_j}}{\rho_0}\right) + \overline{u'_i u'_j}\frac{\partial \overline{u_i}}{\partial x_j} - \frac{g}{\rho_0}\overline{u'_3 \rho'} - 2\nu\overline{s'_{ij} s'_{ij}}$$

Where **turbulent kinetic energy** is

$$\bar{e} = \frac{1}{2}\overline{u'_i u'_i} = \frac{1}{2}\overline{u'^2_1 + u'^2_2 + u'^2_3} = \frac{1}{2}\overline{u'^2 + v'^2 + w'^2}$$

$$\frac{\partial \bar{e}}{\partial t} + \overline{u_j}\frac{\partial \bar{e}}{\partial x_j} = \frac{\partial}{\partial x_j}\left(-\frac{1}{2}\overline{u'_i u'_i u'_j} + 2\nu\overline{u'_i s'_{ij}} - \frac{\overline{p' u'_j}}{\rho_0}\right)$$

$$+ \overline{u'_i u'_j}\frac{\partial \overline{u_i}}{\partial x_j} - \frac{g}{\rho_0}\overline{u'_3 \rho'} - 2\nu\overline{s'_{ij} s'_{ij}}$$

Material derivative = turbulent transport + shear production – buoyancy flux – dissipation

$$\frac{\partial \bar{e}}{\partial t} + \overline{u_j}\cancel{\frac{\partial \bar{e}}{\partial x_j}} = \frac{\partial}{\partial x_j}\left(-\frac{1}{2}\cancel{\overline{u'_i u'_i u'_j}} + \cancel{2\nu\overline{u'_i s'_{ij}}} - \cancel{\frac{\overline{p' u'_j}}{\rho_0}}\right)$$

$$+ \overline{u'_i u'_j}\frac{\partial \overline{u_i}}{\partial x_j} - \frac{g}{\rho_0}\overline{u'_3 \rho'} - 2\nu\overline{s'_{ij} s'_{ij}}$$

Rate of change of tke = shear production – buoyancy flux – dissipation

Because we assume **homogeneous turbulence**:

- ✓ Statistical properties of turbulence are **invariant** in space
- ✓ Equivalently, spatial gradients of turbulent properties vanish

$$\frac{\partial \bar{e}}{\partial t} + \overline{u_j}\cancel{\frac{\partial \bar{e}}{\partial x_j}} = \frac{\partial}{\partial x_j}\left(-\frac{1}{2}\cancel{\overline{u'_i u'_i u'_j}} + 2\nu\cancel{\overline{u'_i s'_{ij}}} - \cancel{\frac{\overline{p' u'_j}}{\rho_0}} \right)$$

$$+ \cancel{\overline{u'_i u'_j}}\frac{\partial \overline{u_i}}{\partial x_j} - \cancel{\frac{g}{\rho_0}\overline{u'_3 \rho'}} - 2\nu \overline{\frac{\partial u'_i}{\partial x_j}\frac{\partial u'_i}{\partial x_j}}$$

Rate of change of tke = – dissipation

Because we assume **homogeneous, isotropic turbulence:**

- ✓ Statistical properties of turbulence are **invariant** in space
- ✓ Equivalently, spatial gradients of turbulent properties vanish
- ✓ No rotation, buoyancy or mean flow

Kolmogorov's hypothesis of local isotropy:

At a sufficiently high Reynolds number, the small scale turbulent motions ($l \ll l_0$) are **locally isotropic**.

Kolmogorov (1941)

Kolmogorov's hypothesis of local isotropy:

At a sufficiently high Reynolds number, the small scale turbulent motions ($l \ll l_0$) are **locally isotropic**.

$$\frac{\partial \bar{e}}{\partial t} = -\varepsilon$$

Kolmogorov (1941)

Kolmogorov's hypothesis of local isotropy:

At a sufficiently high Reynolds number, the small scale turbulent motions ($l \ll l_0$) are **locally isotropic**.

- ✓ **Globally across these scales** the rate of change of $\overline{e}$ is equal and opposite to the rate of its dissipation ε
- ✓ The eddies **"forget" the anisotropy (deformation, rotation, etc)** of large scale motions

Kolmogorov (1941)

Kolmogorov's first similarity hypothesis:

In every turbulent flow at a sufficiently high Reynolds number, the statistics of the small scale motions ($l \ll l_0$) have a universal form that is determined by ν and ε.

Kolmogorov (1941)

Kolmogorov's first similarity hypothesis:

In every turbulent flow at a sufficiently high Reynolds number, the statistics of the small scale motions ($l \ll l_0$) have a universal form that is determined by ν and ε.

- ✓ Energy **cascades** across scales at the dissipation rate ε
- ✓ Once the motions become **small enough**, **molecular viscosity** removes the turbulent kinetic energy from the system

Kolmogorov (1941)

Kolmogorov's second similarity hypothesis:

In every turbulent flow at a sufficiently high Reynolds number, the statistics of the small scale motions at $l_0 \ll l \ll l_k$ have a universal form that is determined by ε, independent of ν.

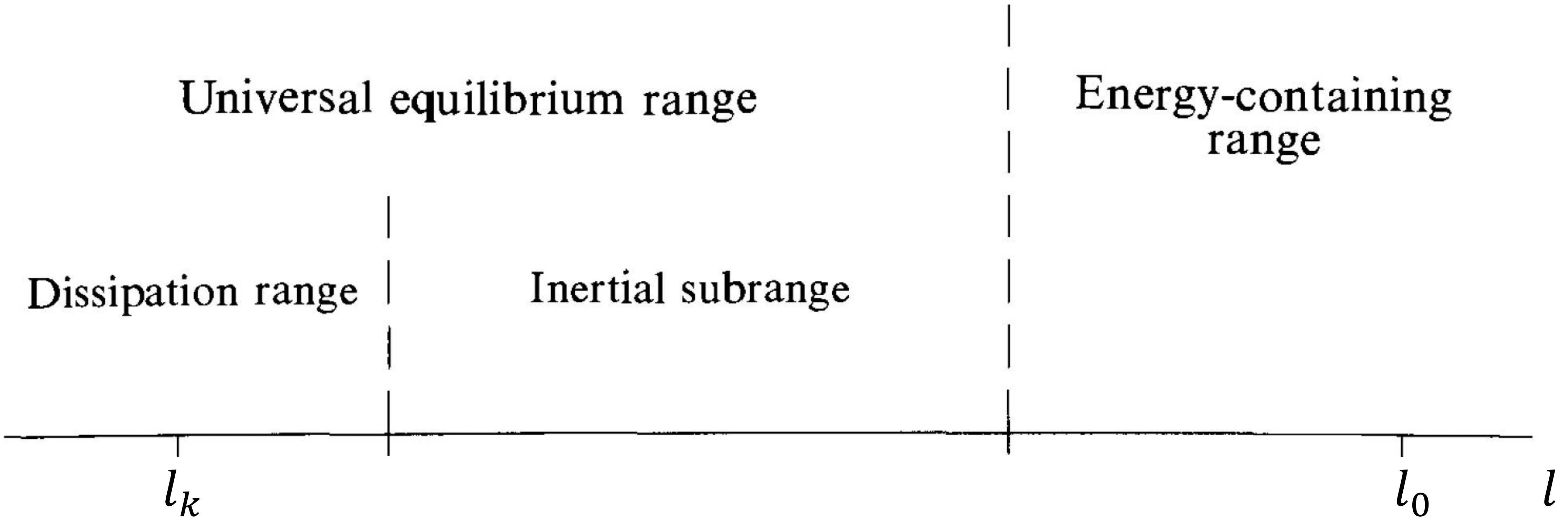
Kolmogorov (1941)

Kolmogorov’s second similarity hypothesis:

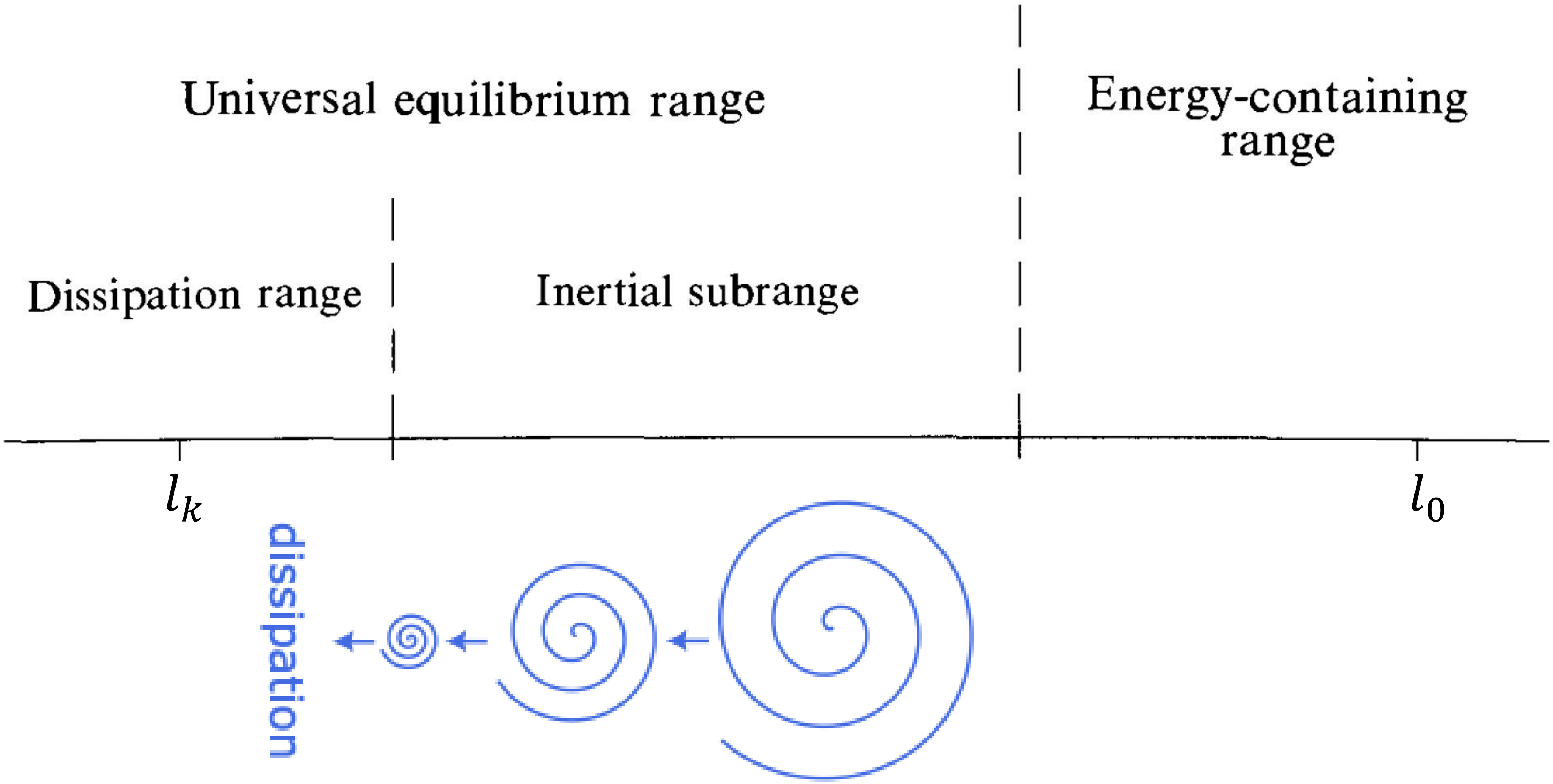
In every turbulent flow at a sufficiently high Reynolds number, the statistics of the small scale motions at $l_0 \ll l \ll l_k$ have a universal form that is determined by ε, independent of ν.

✓ At scales which **“don’t feel”** the effects of molecular viscosity, the energy transfer is **inviscid** and thus independent of ν

Kolmogorov (1941)


Universal equilibrium range
Energy-containing range
Dissipation range
Inertial subrange
l_k
l_0
l

Pope (2000)


Universal equilibrium range
Energy-containing range
Dissipation range
Inertial subrange
l_k
l_0
l
dissipation

Pope (2000)

Summary II:

Kolmogorov's hypotheses *formalise* the previously proposed notion that turbulent kinetic energy is transferred from large to small scales of motion via inviscid processes.

WAVENUMBER SPACE

Energy spectrum: $\overline{e} = \int_0^\infty E(k)dk$

Dissipation spectrum: $\varepsilon = \int_0^\infty D(k)dk$

Spectral energy flux: $\mathcal{T}(k)$

TKE equation in wavenumber space:

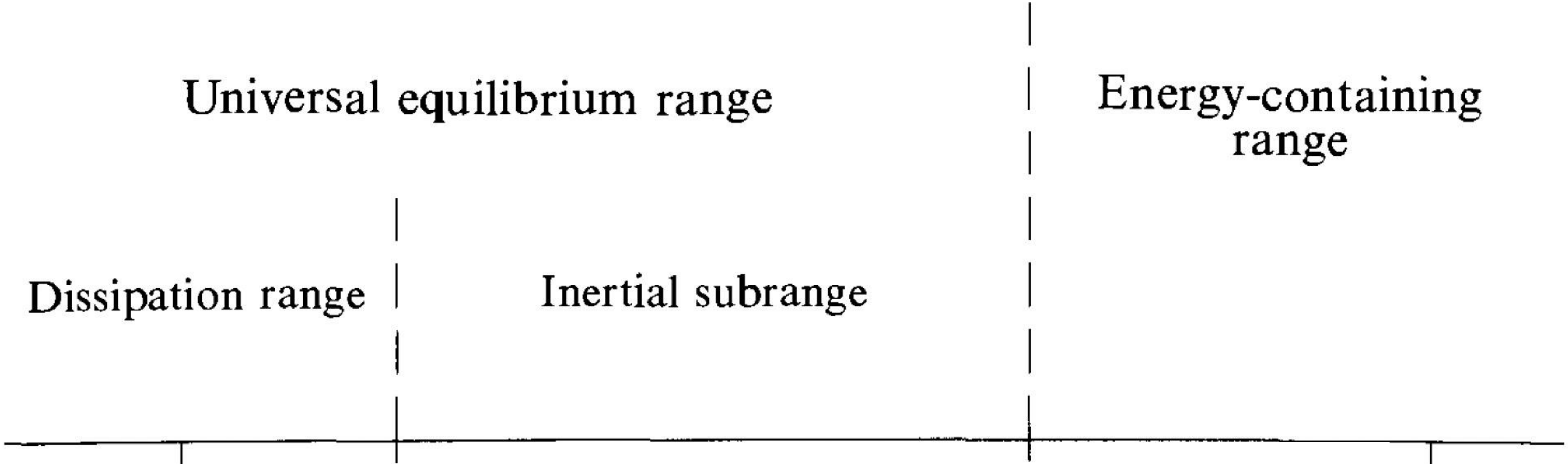
$$\frac{\partial E}{\partial t} - \frac{\partial \mathcal{T}}{\partial k} = -D(k)$$

TKE equation in wavenumber space:

$$\frac{\partial E}{\partial t} - \frac{\partial \mathcal{T}}{\partial k} = -D(k)$$

We can recover $\frac{\partial \overline{e}}{\partial t} = -\varepsilon$ with:

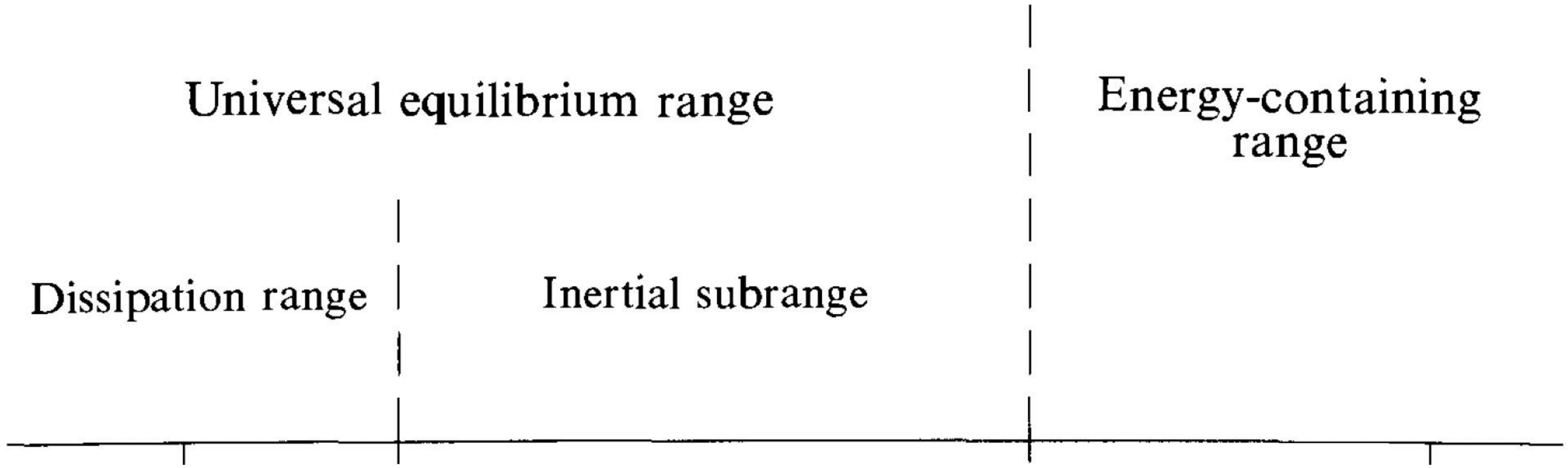
$$\int_0^\infty \frac{\partial E}{\partial t} dk - \int_0^\infty \frac{\partial \mathcal{T}}{\partial k} dk = -\int_0^\infty D(k) dk$$



$k \to \infty$ $\quad k_k \quad$ k_0 $\quad k = 0$

$$\frac{\partial E}{\partial t} = 0 \quad \rightarrow \quad \frac{\partial \mathcal{T}}{\partial k} = D(k)$$

Pope (2000)

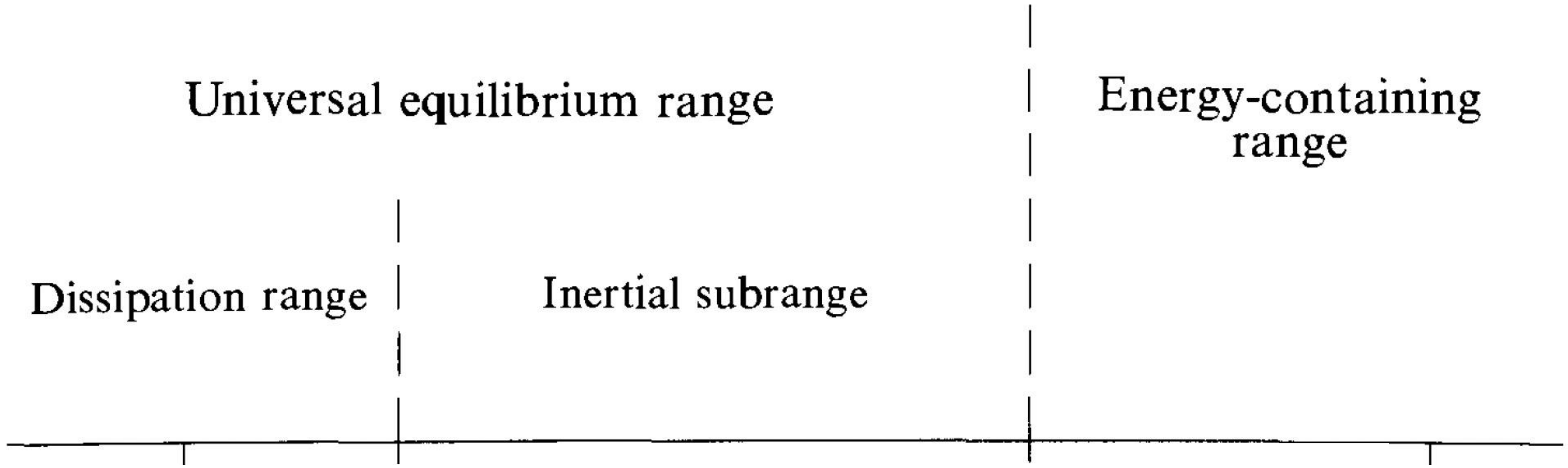


$k \to \infty$ $\quad k_k \quad$ $k_0 \quad k = 0$

$$\frac{\partial E}{\partial t} = 0 \quad \to \quad \frac{\partial \mathcal{T}}{\partial k} = D(k)$$

$$E = E(k, \varepsilon)$$

Pope (2000)



$k \to \infty$ $\quad k_k \quad$ k_0 $\quad k = 0$

$$\frac{\partial E}{\partial t} = 0 \quad \to \quad \frac{\partial \mathcal{T}}{\partial k} = D(k)$$

$$E = E(k, \varepsilon, \nu) \qquad\qquad E = E(k, \varepsilon)$$

Pope (2000)

Summary III:

The transition into wavenumber space allows to *mathematically describe* the process by which turbulent kinetic energy cascades from large scale to small scale motions.

KOLMOGOROV'S SPECTRUM

What do we do when we know the *important parameters* for an equation but don't know its *form*?

What do we do when we know the *important parameters* for an equation but don’t know its *form*?

Dimensional analysis!

Exercise 1: Find a combination of υ and ε that form a length scale, a time scale and a velocity scale.

Exercise 1: Find a combination of υ and ε that form a length scale, a time scale and a velocity scale.

$$[\upsilon] = L^2T^{-1}$$
$$[\varepsilon] = L^2T^{-3}$$

$$[l_k] = [\upsilon^{\alpha}\varepsilon^{\beta}] = L^1T^0$$
$$[t_k] = [\upsilon^{\alpha'}\varepsilon^{\beta'}] = L^0T^1$$
$$[v_k] = [\upsilon^{\alpha''}\varepsilon^{\beta''}] = L^1T^{-1}$$

Conlcusions from Exercise 1

1) The Reynolds number at dissipative scales is

$$Re_k = \frac{l_k v_k}{\nu} = 1$$

2) The dissipative scales are much smaller than scales at which $\overline{e}$ enters the system

$$\frac{l_k}{l_0} \sim Re_0^{-3/4} \ll 1 \quad \frac{t_k}{t_0} \sim Re_0^{-1/2} \ll 1 \quad \frac{v_k}{v_0} \sim Re_0^{-1/4} \ll 1$$

Exercise 2: Find a combination of k and ε that match energy density units.

$[k]$ = $L^{-1}T^{0}$
$[\varepsilon]$ = $L^{2}T^{-3}$
$[E]$ = $L^{3}T^{-2}$

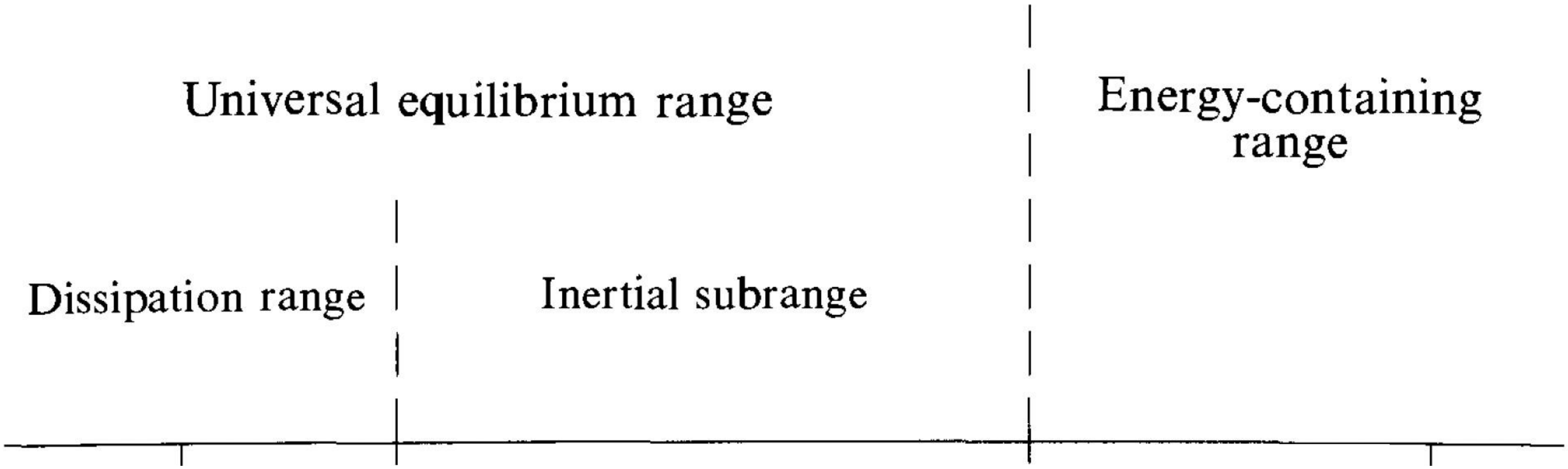
$[E]$ = $[k^{\alpha}\varepsilon^{\beta}]$ =$L^{3}T^{-2}$

Exercise 2: Find a combination of k and ε that match energy density units.

$[k]$ = $L^{-1}T^{0}$
$[\varepsilon]$ = $L^{2}T^{-3}$
$[E]$ = $L^{3}T^{-2}$

$[E]$ = $[k^{\alpha}\varepsilon^{\beta}]$ =$L^{3}T^{-2}$

$$\begin{cases} -\alpha + 2\beta = 3 \\ -3\beta = -2 \end{cases} \quad \Longrightarrow \quad \begin{cases} \alpha = -5/3 \\ \beta = 2/3 \end{cases}$$

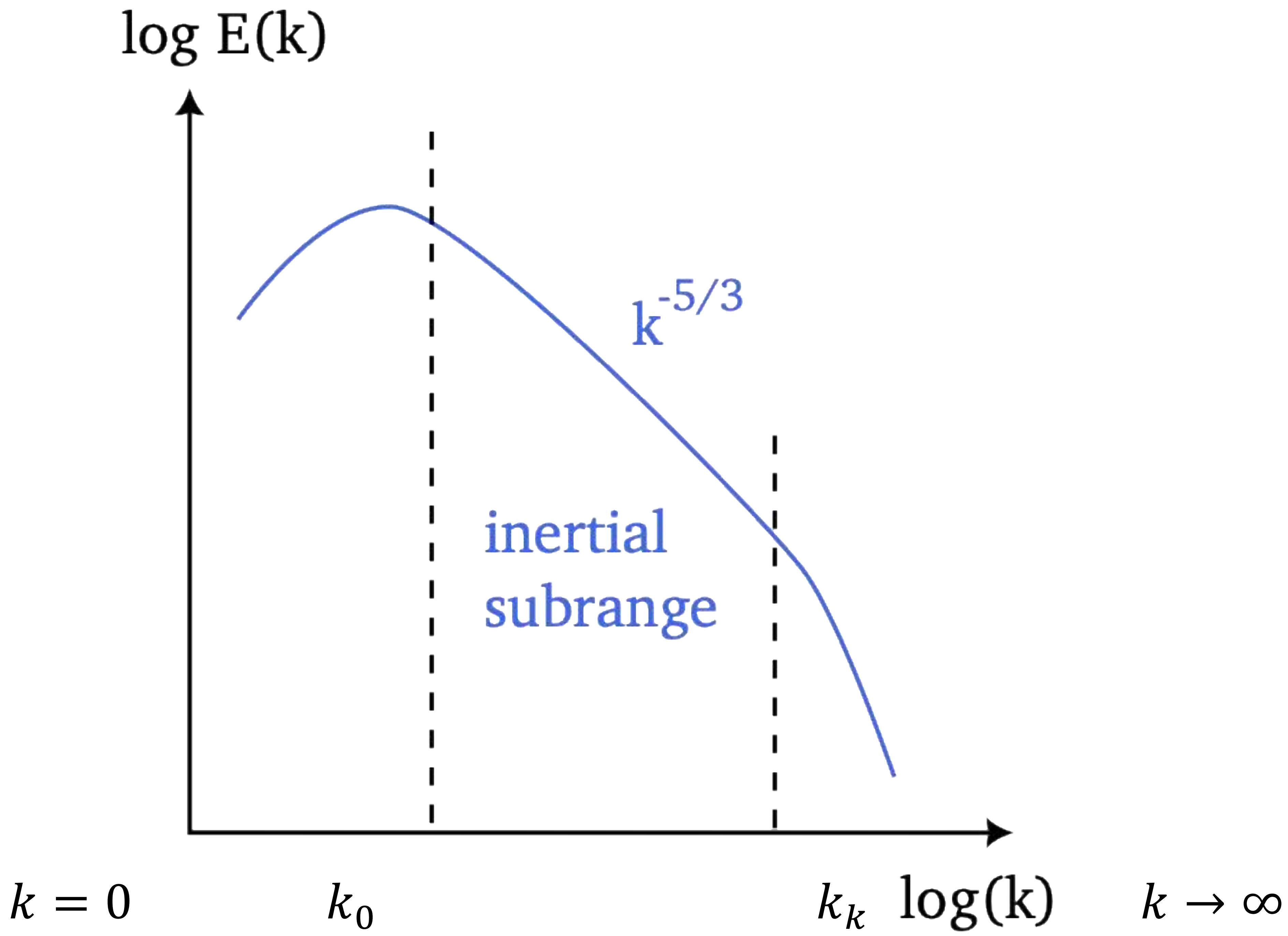


$k \to \infty$ k_k k_0 $k = 0$

$$\frac{\partial E}{\partial t} = 0 \quad \rightarrow \quad \frac{\partial \mathcal{T}}{\partial k} = D(k)$$

$$E = E(k, \varepsilon, \nu) \qquad E(k) \propto \varepsilon^{2/3} k^{-5/3}$$

Pope (2000)

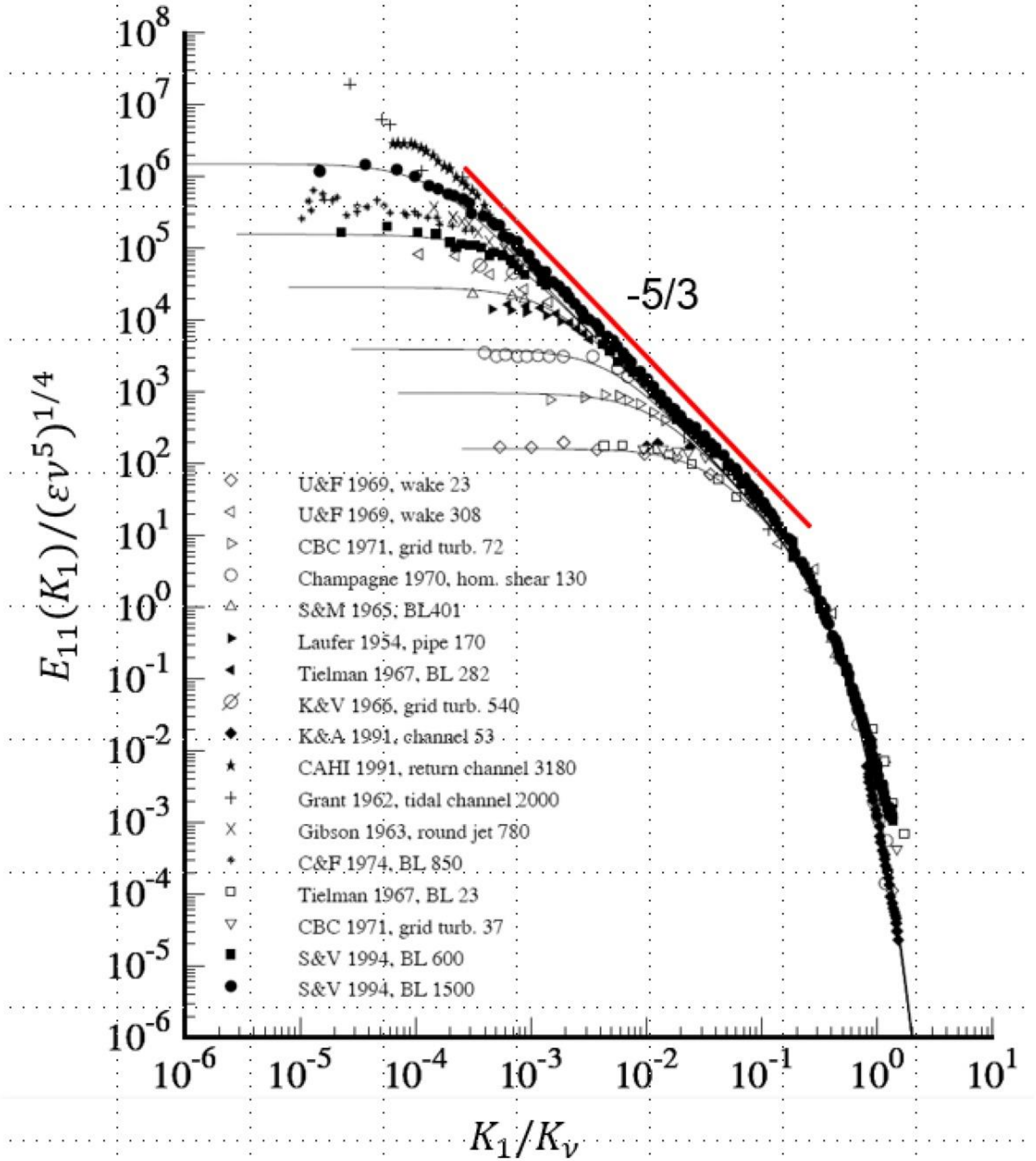

log E(k)
k-5/3
inertial
subrange
k = 0
k0
kk
log(k)
k → ∞

Summary IV:

Deriving the formula for the energy spectrum in the inertial subrange

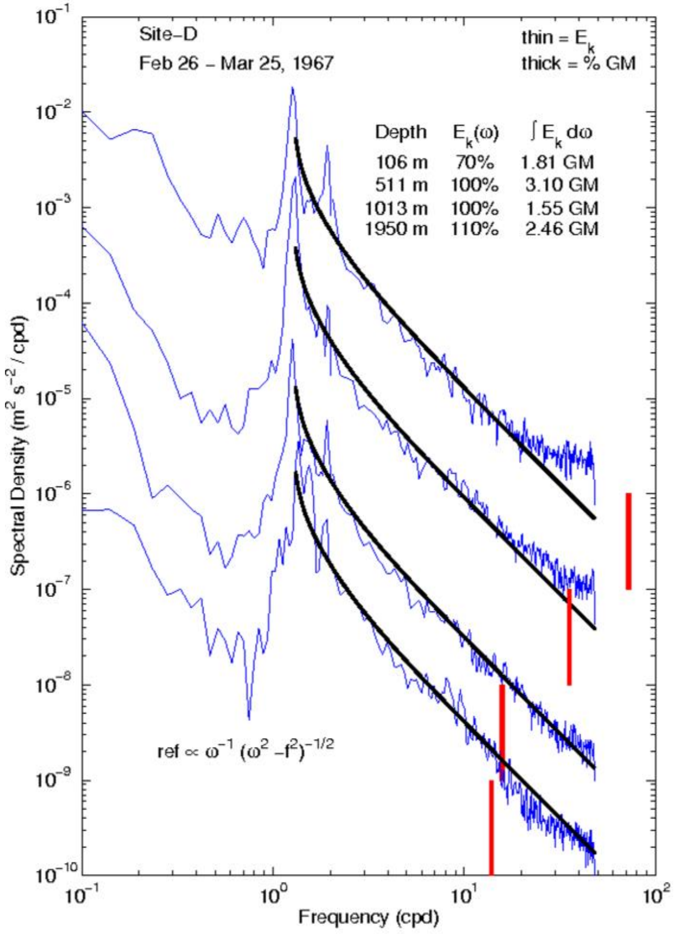
$$E(\varepsilon, k) \propto \varepsilon^{2/3} k^{-5/3} \qquad \text{for} \qquad k_0 \ll k \ll k_k$$

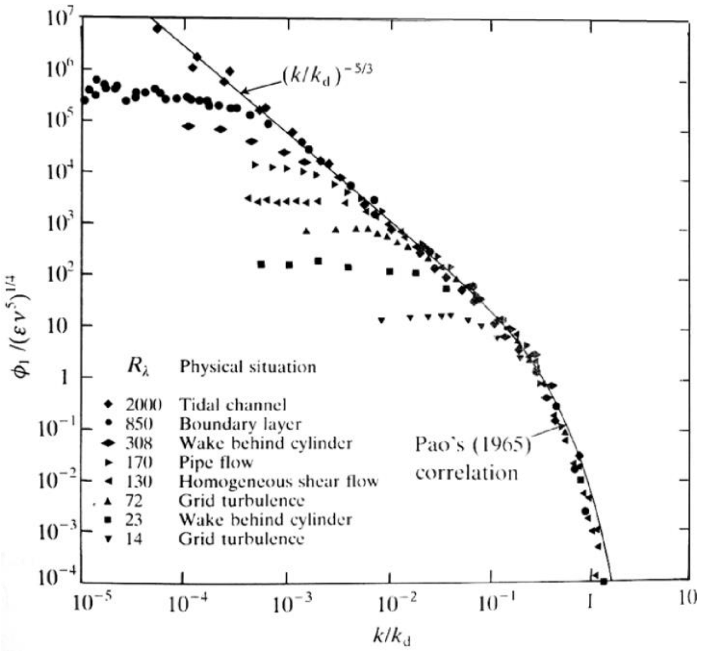
is equivalent to **developing a testable hypothesis for the energy cascade**. It makes the energy cascade itself (indirectly) *observable*.

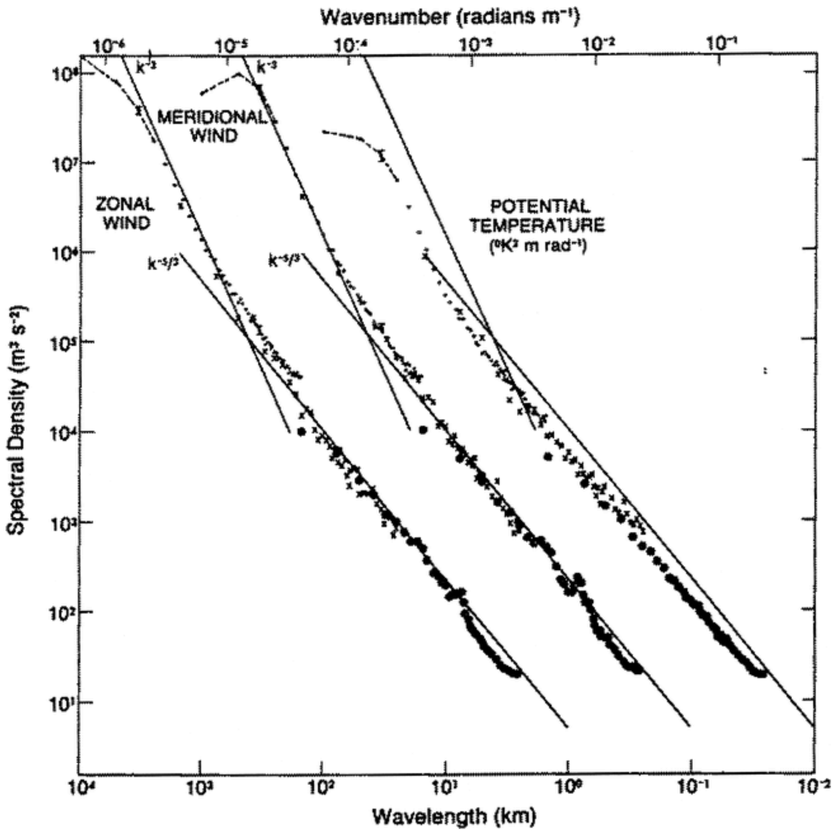


-5/3
U&F 1969, wake 23
U&F 1969, wake 308
CBC 1971, grid turb. 72
Champagne 1970, hom. shear 130
S&M 1965, BL401
Laufer 1954, pipe 170
Tielman 1967, BL 282
K&V 1966, grid turb. 540
K&A 1991, channel 53
CAHI 1991, return channel 3180
Grant 1962, tidal channel 2000
Gibson 1963, round jet 780
C&F 1974, BL 850
Tielman 1967, BL 23
CBC 1971, grid turb. 37
S&V 1994, BL 600
S&V 1994, BL 1500
$E_{11}(K_1)/(\varepsilon\nu^5)^{1/4}$
K_1/K_ν

Pope (2000)







Lvov et al. (2005), McComb and May (2004), Nastrom and Gage (1982)

Resources

My favorite textbook on turbulence:
"Turbulent Flows", Stephen B. Pope (2000)

Fantastic notes on marine turbulence:
https://www.io-warnemuende.de/files/staff/umlauf/turbulence/turbulence.pdf

Signal analysis in oceanography (includes beautiful data visualisation and encouraging words):
https://www.jmlilly.net/index.html

You are welcome to request papers included in this presentation from me:
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